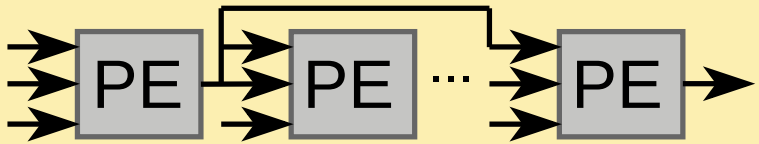
Single IMC Primitive

Multi-Primitive IMC

Homogeneous Workload (same operation type)

ISAAC [ISCA '16]
NeurRRAM [Nature '22]
PRIME [ISCA '16]

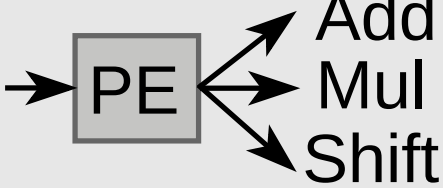
Established IMC paradigm



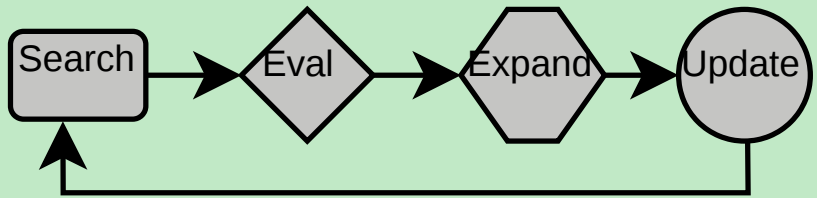
HEIRS [DAC '24]
AIG-CIM [DAC '24]

Multiple memories, same operation (neural inference)

Heterogeneous Workload (different operations per phase)



One primitive has to serve multiple distinct workloads



New: multi-primitive IMC for phase-heterogeneous workloads (neurosymbolic workloads)